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Housing Prices: Inference and Prediction Using Regression Methods

Microeconometrics Project

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Abstract

Housing prices play an important role in household welfare and financial decision-making. Understanding the factors that determine housing values is therefore essential for both buyers and policymakers. This paper analyzes housing prices using a cross-sectional dataset of 545 residential properties obtained from Kaggle and evaluates several regression-based approaches for explanation and prediction.

A multiple linear regression model with heteroskedasticity-robust standard errors is estimated to assess the significance of housing characteristics. Model selection is performed using best subset selection based on adjusted R^2. For prediction, ridge regression is applied with the penalty parameter selected through cross-validation, and predictive performance is evaluated using a separate test sample.

The results indicate that housing prices are strongly associated with property size, amenities, and location characteristics, while the number of bedrooms has limited explanatory power once total area is controlled for. Ridge regression provides a modest improvement in predictive accuracy relative to alternative models. Overall, the findings highlight the importance of distinguishing between inference and prediction when modeling housing prices.

1 Introduction

Housing plays a crucial role in both individual well-being and household financial security. The cost, availability, and quality of housing directly influence living standards and household budgets, making housing decisions an important economic concern.

Housing prices are shaped by a combination of factors, including the size of the property, available amenities, and its location. Economists often use regression-based methods to estimate how each of these characteristics affects property values. However, models designed mainly for statistical analysis do not always excel at predicting prices, especially when the factors are closely related.

This paper examines housing prices using a cross-sectional dataset of 545 residential properties. It focuses on two key questions:
(i) which housing characteristics drive price differences.
(ii) which modeling approach yields the most accurate predictions.
By separating the goals of explanation and prediction, the study shows that models that help us understand housing markets are not always the best at forecasting prices.

The paper is structured as follows. Section 2 introduces the data, Section 3 explains the methodology, Section 4 presents the results, and Section 5 offers concluding insights.

2 Data Description

Although the dataset does not specify a location, it could theoretically represent housing data from India, possibly from urban residential areas along major roads (e.g., MG Road or sector-based neighborhoods). This assumption is based on common real estate terminology and dataset usage. It contains 545 observations and 13 variables describing house prices and property characteristics. The dependent variable is **price**, measured in Indian Rupees . Explanatory variables include the **area** of the house measured in square feet, the number of **bedrooms**, **bathrooms**, **stories** (floors), and **parking** spaces, all expressed as counts.

The dataset also contains several binary variables indicating housing amenities, including **mainroad** (whether the house is connected to a main road), **guestroom** (presence of a guest room), **basement**, **hotwaterheating**, **airconditioning**, and **prefarea** (whether the house is located in a preferred area).

Additionally, the variable **furnishingstatus** describes the furnishing level of the property and takes three categories: *furnished*, *semi-furnished*, and *unfurnished*.

Before analysis, the data were checked for missing values and duplicate records, and none were found, indicating that the dataset was clean and suitable for analysis.

3 Descriptive analysis

3.1 Mean and Standard Deviation

The mean house price is 4,766,729 INR which represents the average price of the houses in the dataset. This gives a general idea of the overall price level in the market.

The standard deviation is 1,870,440 INR which indicates that house prices tend to vary quite a lot around the mean. In other words, prices are fairly spread out rather than concentrated near the average.

3.2 Distribution of House Prices

Figure 1 shows house prices ranging from 2,000,000 to 13,000,000 Indian Rupees. Most homes fall between 3,000,000 and 6,000,000 Indian Rupees, while a few very expensive properties stretch the distribution to the right. This suggests that mid-range homes are the norm for most buyers, and the few high-end properties push up the average price. This indicates that high-priced homes are uncommon but have a strong influence on the overall price pattern.

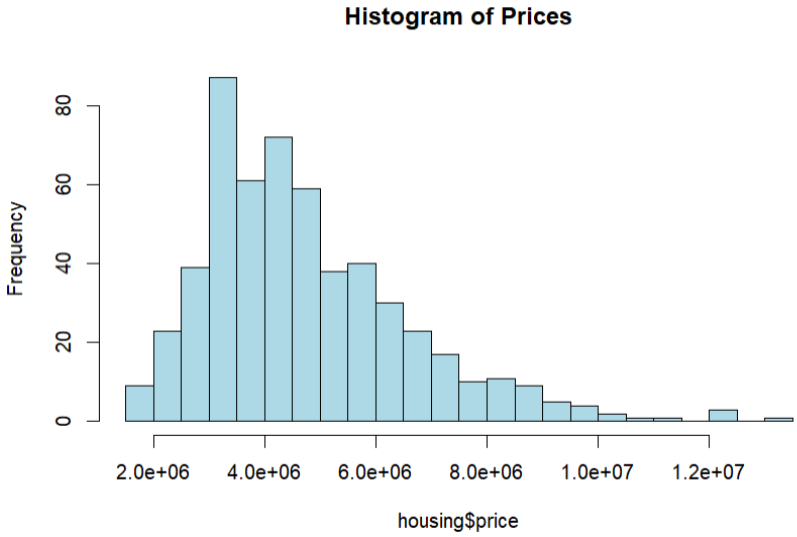


Figure 1: Histogram of house prices.

3.3 Box Plot of House Prices

The box plot in Figure 2 divides the data into quartiles. It shows that about "25% of houses cost less than around 3 million (Q1), about 50% cost less than around 4 million INR (median), and about 75% cost less than around 6 million INR (Q3). This means that the middle 50% of house prices lie between approximately 3 million INR and 6 million INR, while a few high-value outliers represent unusually expensive homes.

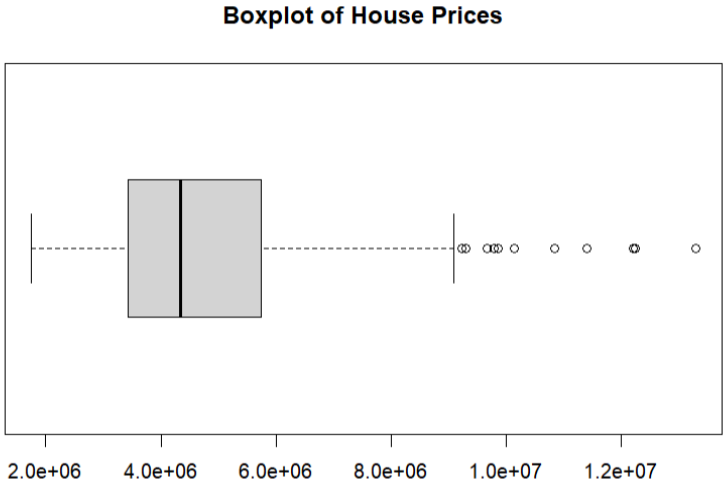


Figure 2: Box plot of house prices.

3.4 Relationship Between Area and Price

The scatterplot of area versus price shown in Figure 3 indicates a positive relationship between house size and price. In general, larger houses tend to be more expensive, although the spread of points shows that area alone does not fully explain price differences. Other factors such as location, design, and amenities may also affect housing values.

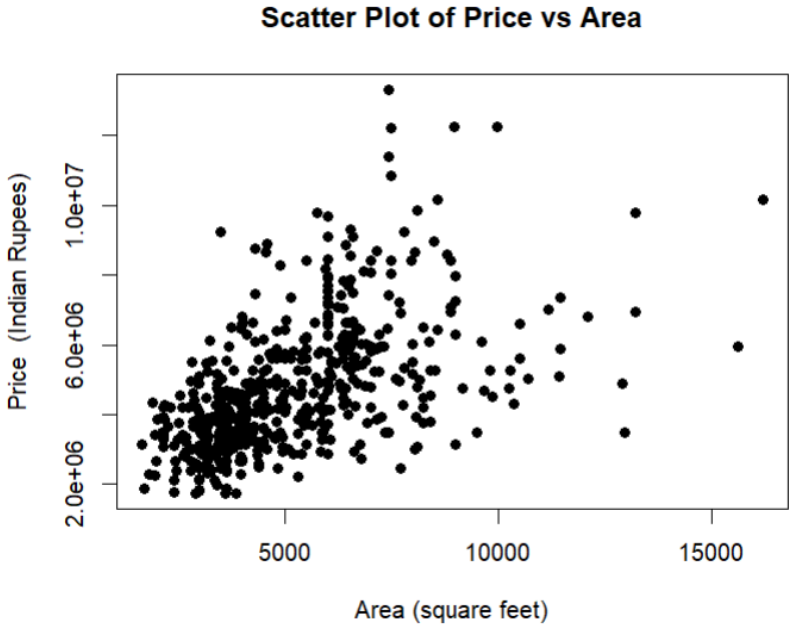


Figure 3: Scatterplot of price versus house area.

3.5 Summary Statistics

The summary statistics shown in Table 1 confirm the wide range of values in the dataset. House prices range from low-cost properties to high-value luxury homes, while house areas also vary considerably. This demonstrates that the dataset contains substantial variability, making it suitable for meaningful statistical analysis.

Table 1: Summary Statistics for Numeric Variables

Variable	Min	Q1	Median	Mean	Q3	Max
Price	1,750,000	3,430,000	4,340,000	4,766,729	5,740,000	13,300,000
Area	1,650	3,600	4,600	5,151	6,360	16,200
Bedrooms	1	2	3	2.965	3	6
Bathrooms	1	1	1	1.286	2	4
Stories	1	1	2	1.806	2	4
Parking	0	0	0	0.694	1	3

3.6 Correlation

As illustrated in the heatmap (see Fig 4), strong correlations are observed between price and key structural features such as area, number of bathrooms, and number of stories, while bedrooms and bathrooms are also closely related. Moderate correlations indicate that house size (area) has a noticeable influence on the number of bedrooms, bathrooms, and stories, and that bedrooms moderately affect price. In contrast, weak correlations show that parking has little relationship with most variables and minimal impact on house price.

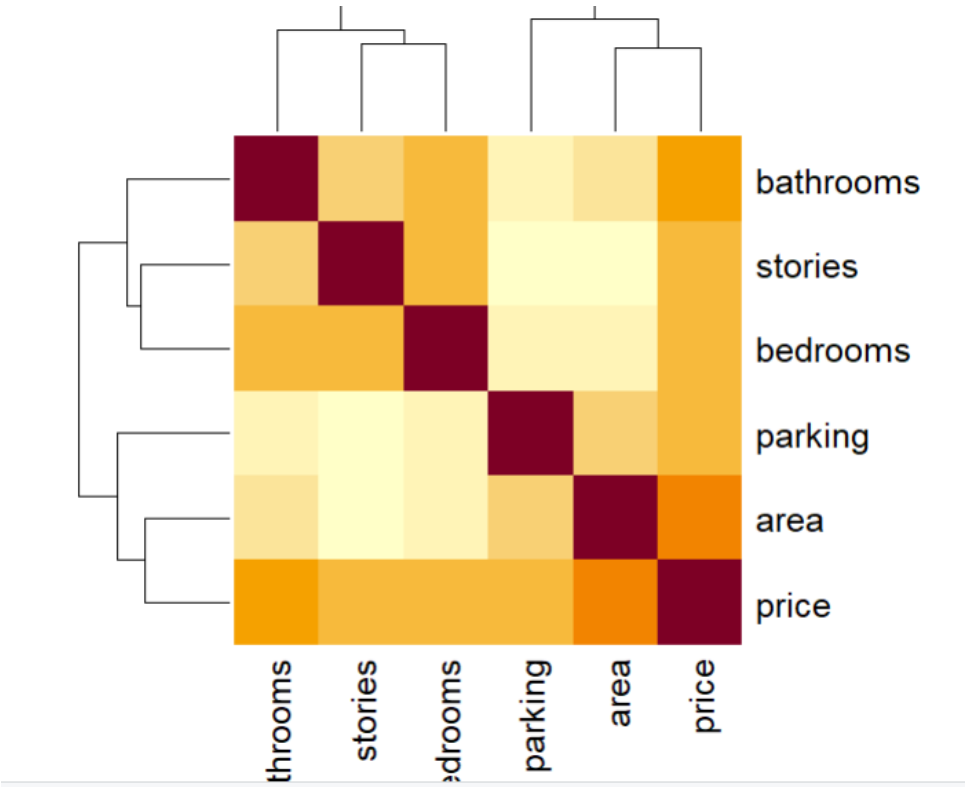


Figure 4: Correlation heatmap of the housing dataset variables

4 Methods

This section describes the empirical strategy used to analyze the determinants of housing prices and to compare the predictive performance of alternative models. The analysis distinguishes between two main objectives: (i) inference on which factors significantly affect housing prices, and (ii) prediction of prices for unseen properties. All models are estimated on a training sample, while predictive accuracy is evaluated on a separate test sample.

4.1 Train–Test Split

To evaluate out-of-sample performance, the data are split into 80% for training and 20% for testing. The training set is used for model fitting and selection, and the test set provides an unbiased measure of predictive accuracy. After splitting the dataset, 436 observations are used for model estimation, while the remaining data are reserved for out-of-sample testing.

4.2 Baseline Regression Model

The starting point of the analysis is a multiple linear regression model of the form

$$\begin{aligned}\text{price}_i = \beta_0 &+ \beta_1 \text{area}_i + \beta_2 \text{bedrooms}_i + \beta_3 \text{bathrooms}_i + \beta_4 \text{stories}_i \\ &+ \beta_5 \text{mainroad}_i + \beta_6 \text{guestroom}_i + \beta_7 \text{basement}_i + \beta_8 \text{hotwaterheating}_i \\ &+ \beta_9 \text{airconditioning}_i + \beta_{10} \text{parking}_i + \beta_{11} \text{prefarea}_i + \beta_{12} \text{furnishingstatus}_i + \varepsilon_i \end{aligned} \tag{1}$$

where $price_i$ is the transaction price of house i; $area_i$, $bedrooms_i$, $bathrooms_i$, $stories_i$, and $parking_i$ measure structural characteristics; the binary variables $mainroad_i$, $guestroom_i$, $basement_i$, $hotwaterheating_i$, $airconditioning_i$, and $prefarea_i$ indicate housing amenities and location; $furnishingstatus_i$ captures the furnishing condition; and ε_i represents unobserved determinants of price.

4.3 Heteroskedasticity-Robust Inference

Residual plots suggest the presence of heteroskedasticity. Therefore, the model is re-estimated using heteroskedasticity-robust standard errors to ensure valid inference. The estimated coefficients and their significance remain stable, indicating that the results are not driven by non-constant variance or outliers.

4.4 Best Subset Selection (Prediction)

Best subset selection is applied to the training data as an exploratory model selection procedure. All possible subsets of explanatory variables are evaluated, and the preferred specification is chosen based on adjusted R^2, which balances model fit and complexity.

4.5 Ridge Regression (Prediction)

For predictive analysis, ridge regression is implemented as a regularized alternative to OLS. Ridge regression minimizes the following objective function:

$$\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (price_i - \mathbf{x}_i'\boldsymbol{\beta})^2 + \lambda \sum_{j=1}^{p} \beta_j^2, \tag{2}$$

where $\lambda \geq 0$ is the regularization parameter controlling the degree of shrinkage. The penalty parameter is selected via K-fold cross-validation using the training data by minimizing the average mean squared error.

Ridge regression is employed exclusively for prediction, as coefficient shrinkage introduces bias that limits interpretability.

4.6 Evaluation Criteria

Two distinct evaluation frameworks are used.

Inference. Inference is based on coefficient estimates, heteroskedasticity-robust standard errors, t-statistics, p-values, confidence intervals, and adjusted R^2.

Prediction. Predictive performance is evaluated on the test sample using mean squared error (MSE), root mean squared error (RMSE). Test-set R^2 measures the proportion of price variation explained in unseen data and provides an indicator of model generalization performance. Since an explicit test set is available, final model selection is based on test-set RMSE. Cross-validation is used exclusively to tune the ridge regularization parameter.

5 Main Results

5.1 Determinants of Housing Prices

Table 2 reports the baseline OLS estimates together with heteroskedasticity-robust standard errors. The model explains approximately 72% of the variation in housing prices with an adjusted R^2 of about 0.71, indicating strong explanatory power.

The estimated pricing equation from the OLS model is:

$$\begin{aligned}\hat{price}_i = {} & 88{,}802 + 231.24\, area_i + 67{,}705\, bedrooms_i + 1{,}001{,}184\, bathrooms_i + 509{,}648\, stories_i \\ & + 424{,}567\, mainroad_i + 337{,}211\, guestroom_i + 368{,}302\, basement_i \\ & + 1{,}111{,}131\, hotwaterheating_i + 880{,}686\, airconditioning_i + 294{,}762\, parking_i \\ & + 613{,}949\, prefarea_i - 57{,}814\, semi_furnished_i - 405{,}483\, unfurnished_i \end{aligned} \tag{3}$$

Housing prices are primarily driven by size, structural quality, and amenities. Floor area, bathrooms, and stories all exhibit strong positive effects, reflecting the economic valuation of space and housing quality. Hot water heating and air conditioning generate especially large premia, highlighting the importance of comfort and infrastructure. Location and accessibility also matter, as properties in preferred areas and near the main road command higher prices. Unfurnished houses sell at a significant discount. The number of bedrooms is statistically insignificant once floor area and quality measures are controlled for, indicating that space matters more than room count.

A Breusch–Pagan test reveals heteroskedasticity (p-value $= 1.51 \times 10^{-7}$), so robust standard errors are used to ensure reliable inference.

Re-estimating the model with heteroskedasticity-robust standard errors yields nearly identical coefficients and significance levels, confirming that the results are not driven by heteroskedasticity.

Table 2: OLS and Robust OLS estimates

Variable	OLS Coef.	Robust SE	p-value
Area	231.24	29.78	< 0.001
Bedrooms	67,705	76,888	0.379
Bathrooms	1,001,184	139,574	< 0.001
Stories	509,648	66,975	< 0.001
Main road (yes)	424,567	116,132	< 0.001
Guest room (yes)	337,211	154,251	0.029
Basement (yes)	368,302	130,590	0.005
Hot water heating (yes)	1,111,131	296,742	< 0.001
Air conditioning (yes)	880,686	120,810	< 0.001
Parking	294,762	70,821	< 0.001
Preferred area (yes)	613,949	134,596	< 0.001
Semi-furnished	-57,814	131,694	0.662
Unfurnished	-405,483	128,272	0.002
Adj R^2		0.709	
Observations		436	

5.2 Predictive Performance

Best subset selection based on adjusted R^2 yields a model with **11 predictors**, excluding only the number of bedrooms. This indicates that bedroom count adds little explanatory power after controlling for size and amenities.

The estimated best subset equation is:

$$\begin{aligned} \hat{price}_i = {} & 226{,}521 + 232.81\, area_i + 1{,}023{,}471\, bathrooms_i + 527{,}799\, stories_i \\ & + 298{,}356\, parking_i + 407{,}670\, mainroad_i + 332{,}796\, guestroom_i \\ & + 377{,}107\, basement_i + 1{,}122{,}172\, hotwaterheating_i + 883{,}070\, airconditioning_i \\ & + 618{,}865\, prefarea_i - 55{,}654\, semi_furnished_i - 407{,}495\, unfurnished_i \end{aligned} \tag{4}$$

Table 3: Best subset model (Adjusted R^2)

Variable	Coefficient	p-value
Area	232.81	< 0.001
Bathrooms	1,023,471	< 0.001
Stories	527,799	< 0.001
Parking	298,356	< 0.001
Main road (yes)	407,670	0.005
Guest room (yes)	332,796	0.013
Basement (yes)	377,107	0.001
Hot water heating (yes)	1,122,172	< 0.001
Air conditioning (yes)	883,070	< 0.001
Preferred area (yes)	618,865	< 0.001
Unfurnished	-407,495	0.002
Adj R^2	0.709	

Table 4 summarizes predictive accuracy on the test sample.

Table 4: Test-set predictive performance

Model	MSE	RMSE	Test R^2
OLS (full)	1.69×10^{12}	1,298,270	0.532
Best subset	1.70×10^{12}	1,303,730	0.528
Ridge (CV)	1.67×10^{12}	1,292,225	0.537

Ridge regression obtains the lowest RMSE and the highest test R^2, meaning that it gives the most accurate predictions for new houses in the dataset. Although the numerical differences with OLS and the best subset model are small, the results are consistent across all measures, which strengthens confidence in this finding.

From an economic point of view, this implies that housing prices are influenced by many related characteristics at the same time, such as size, location, and amenities. Shrinking coefficients helps the model combine this information more effectively, leading to more stable and reliable price predictions. While OLS and best subset models are useful for understanding how each feature affects price, ridge regression provides better guidance when the goal is to predict market values.

6 Conclusion

Results show that prices are mainly driven by size, quality, and comfort. Larger houses, more bathrooms and stories, and amenities such as air conditioning, heating, and basements increase prices. Location also matters, as properties in preferred areas or near main roads are more expensive. In contrast, the number of bedrooms is not significant once total area is included, suggesting that overall space is more important than room count.

The use of robust standard errors confirms the reliability of these results. For prediction, ridge regression slightly outperforms OLS and best subset models by producing lower prediction errors, indicating gains from coefficient stabilization rather than variable selection.

These results leave room for further research, particularly by incorporating richer neighborhood data or exploring alternative modeling approaches to better capture housing price dynamics.

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